PATRICK YIN

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EDUCATION

University of Washington 2023–
Ph.D. Computer Science & Engineering
Awards: NSF Graduate Research Fellow, ∼ 10% selection rate

University of California, Berkeley 2019–2023
B.A. Computer Science, GPA: 4.0/4.0
Awards: Regents' and Chancellor's Scholar, top <2% incoming class; National Merit Scholar

EXPERIENCE

Research Intern, NVIDIA — *Advised by Ankur Handa, Karl Van Wyk, Nathan Ratliff* 2026–
Working on reinforcement learning and sim2real transfer for dexterous hands.

Graduate Researcher, UW Robotics — *Advised by Abhishek Gupta* 2023–
Working on reinforcement learning and sim2real transfer for dexterous robot manipulation.

Student Researcher, Microsoft Research — *Advised by Andrey Kolobov* 2025–2026
Research Intern, Microsoft Research 2025
Worked on reinforcement learning and sim2real transfer for dexterous robot manipulation.

Undergrad Researcher, Berkeley AI Research — *Advised by Sergey Levine, Kuan Fang, Ashvin Nair* 2020–2023
Published 4 papers on offline goal-conditioned reinforcement learning and finetuning for real-world robotic control.

Machine Learning Engineer Intern, Ambi Robotics — *Advised by Yi Li* 2022
Spearheaded training deep learning models on real-world production data and training 3D neural networks.

Software Engineer Intern, UiPath 2021
Pushed over 30 Git commits to production on Insights team working with Snowflake/SQL, Kubernetes, and Docker.

SELECTED PUBLICATIONS

AlphaNIST: Emergent Assembly Skills from Binary Task Rewards
O. Zhang, I. Akinola, R. Scalise, M. Guaman Castro, **P. Yin**, T. Widmer, M. Macklin, C. Amevor, D. Turpin, D. Van Gelder, J. Carius, A. Handa, G. State, Y. Guo, Y. Narang
In submission

A Balanced Data Diet: Addressing the Bottleneck in Mega-Scale RL for Robot Control
O Zhang*, M Castro*, **P Yin***, I Dagnino, A Gupta, R Scalise, B Boots
Conference on Robot Learning (CoRL), 2026

OmniReset: Emergent Dexterity Via Diverse Resets and Large-Scale Reinforcement Learning
P Yin*, T Westenbroek*, O Zhang, J Tran, I Dagnino, E Shilamkar, N Tiapo, S Bagaria, X Liu, G Mullins, A Kolobov, A Gupta
International Conference on Learning Representations (ICLR), 2026

OTHER PUBLICATIONS

Visual Sim-to-Real Learning for Robotic Insertion under Geometric Variations: Rebar Installation
T Sun*, B Han*, **P Yin**, R Xu, H He, A Gupta, S Rusinkiewicz, Y Shao
In Submission

Simulation Distillation: Pretraining World Models in Simulation for Rapid Real-World Adaptation
J Levy*, T Westenbroek*, K Huang, F Palafox, **P Yin**, S Omidshafiei, D Kim, A Gupta, D Fridovich-Keil
Robotics: Science and Systems (RSS), 2026

SGFT: Rapidly Adapting Policies to the Real-World via Simulation-Guided Fine-Tuning
P Yin*, T Westenbroek*, S Bagaria, K Huang, C Cheng, A Kolobov, A Gupta
International Conference on Learning Representations (ICLR), 2025

DROID: A Large-Scale In-the-Wild Robot Manipulation Dataset
A Khazatsky*, K Pertsch*, ..., **P Yin**, et al.
Robotics: Science and Systems (RSS), 2024

ASID: Active Exploration for System Identification and Reconstruction in Robotic Manipulation
M Memmel, C Zhu, A Wagenmaker, **P Yin**, D Fox, A Gupta
International Conference on Learning Representations (ICLR), 2024 (Oral Presentation)

Stabilizing Contrastive RL: Techniques for Robotic Goal Reaching from Offline Data
C Zheng, B Eysenbach, H Walke, **P Yin**, K Fang, R Salakhutdinov, S Levine
International Conference on Learning Representations (ICLR), 2024 (Spotlight Talk)

Open X-Embodiment: Robotic Learning Datasets and RT-X Models
Open X-Embodiment Collaboration, ..., **P Yin**, et al.
IEEE International Conference on Robotics and Automation (ICRA), 2024 (Best Paper)

Generalization with Lossy Affordances: Leveraging Broad Offline Data for Learning Visuomotor Tasks
K Fang, **P Yin**, A Nair, H Walke, G Yan, S Levine
Conference on Robot Learning (CoRL), 2022 (Oral Presentation)

Planning to Practice: Efficient Online Fine-Tuning by Composing Goals in Latent Space
K Fang*, **P Yin***, A Nair, S Levine
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022

Bisimulation Makes Analogies in Goal-Conditioned Reinforcement Learning
P Hansen-Estruch, A Zhang, A Nair, **P Yin**, S Levine
International Conference on Machine Learning (ICML), 2022

AWARDS

NSF Graduate Research Fellowship. $\sim$ 10% selection rate. 2025
Highest Distinction in General Scholarship, L&S CS. Top $\sim$ 3% of graduating class at Cal. 2023
Regents' and Chancellor's Scholar. Top $<$ 2% of incoming class at Cal. 2019
National Merit Scholar. Top 0.02% of high school seniors in US. 2019

MENTORSHIP

Yanda Bao — BS @ UW
Joshua Tran — BS @ UW
Ignacio Dagnino — Student @ Skyline High School $\rightarrow$ Researcher @ UW
Simran Bagaria — BS/MS @ UW $\rightarrow$ Research Engineer @ Microsoft Research $\rightarrow$ Research Engineer @ Eka Robotics

PROFESSIONAL SERVICE

Teaching Assistant
CSE 391: System and Software Tools 2026
CSE 542: Reinforcement Learning 2024

Reviewer
IEEE Transactions on Robot Learning (T-RL) 2026
IEEE International Conference on Robotics and Automation (ICRA) 2025,2026
Computer Vision and Pattern Recognition Conference (CVPR) 2024,2025
Conference on Robot Learning (CoRL) 2024

UW CSE PhD Admissions Reviewer 2024